
Introduction

Transdisciplinary education involves two types of environments:

- (The educational diversity of applied sciences)
- Professional field context, involving partners such as the

This includes both the *soft side* of collaboration and the *physical locations* where the work takes place. It's important to determine how these contexts are connected within the transdisciplinary module. This influences what and how people learn and work in transdisciplinary education.

On the **Theory** page, you'll find explanations of how to understand these environments and the forms they can take.

Under the **Practice** tab, you'll find examples of existing projects and how they've designed their learning environments.

Theory

The term *learning environment* refers to the context in which different partners collaborate. This also translates into the physical locations where module participants learn and work. The types of learning environments below help clarify this. This overview presents a **continuum** — a seamless line with the educational context on one end and the work/societal context on the other. Within this continuum, the following learning environments can be distinguished (Bouw, 2020; Bos et al., 2024):

1. Learning environments based on alignment

Learners move back and forth between the separate contexts of education and external environments.

Targeted (group) interventions such as preparation sessions, return days, and coaching conversations help align education with the external context.

2. Learning environments based on incorporation

Part of the educational context is incorporated into the external context, or vice versa.

For example, aspects can be practiced in a simulated setting at the educational location.

In the external context, learners might receive just-in-time theoretical instruction or guided practice with skills.

3. Learning environments based on hybridization

Parts of the educational and external contexts merge to create a new practice, with characteristics of both.

Physical Location

Conducting transdisciplinary education within the educational context has the advantage of well-equipped spaces for work, learning, and collaboration — think projectors, screens, tables, chairs, and Wi-Fi. Moreover, the campus is accessible and familiar to students, which can lower the threshold for participation. For societal partners, however, the

campus may lie outside their comfort zone.

Choosing an **external location** can be inspiring and may expose existing complex challenges that participants can address. This setting offers space for experimentation, research, and collaboration — often with societal partners already involved at the location. It also breaks the routine of a familiar environment, creating a **neutral space** where all participants engage on equal footing (Wagemakers et al., 2024).

It's crucial to consider the context:

What kind of environment do you want to create with the participants?
Should it be informal, practical, or social?
The choice depends on the complex issue at hand.

At **Avans University of Applied Sciences**, for example, **HELIOs** (Hybrid External Learning and Innovation Environments) go beyond the classroom.

- **The Klein-Ardevedig garden** focuses on issues like food, climate, and diversity.
- **The High West business incubator** may be as much as

References:

Bouw, E., & Zitter, I. (2020). Ontwerppadwegingen voor leeromgevingen op de grens van school en werk: verbinden van school en werk. Canon beroepsonderwijs mei 2020.

Wagemakers, S., Jacobs, L., Kesselring, M., Rosier, B., Schenkels, A., van Teeffelen, S., Tilma, I., & Wolfensberger, M. (2024). *Tijd voor transdisciplinair samenwerken: HELIO's in de praktijk* [magazine]. Avans Hogeschool.

Practice

The **Hybrid Learning Environment Framework** can be used as a discussion tool when developing a hybrid transdisciplinary learning environment.

[Werkkaart bij Raamwerk Leerwerkomgevingen AANGEPAST.pptx \(sharepoint.com\)](#) (only Avans intern)

Case Example: HELIOs

At **Avans University of Applied Sciences**, the concept of **HELIOs** is used: **Hybrid External Learning and Innovation**

Environments. These

are physical spaces outside the walls of educational institutions where education, research, the professional field, and the surrounding community (e.g., neighborhood residents) come together.

These

environments facilitate collaboration and bring together different perspectives to address societal challenges. They are places to learn, research, and create together with others.

More information and examples of HELIOs can be found in [this magazine](#).